

PRINCIPLE OF MATHEMATICAL INDUCTION:

Let $P(n)$ be a propositional function defined for all positive integers n . $P(n)$ is true for every positive integer n if

1. Basis Step:

The proposition $P(1)$ is true.

2. Inductive Step:

If $P(k)$ is true then $P(k + 1)$ is true for all integers $k \geq 1$.

i.e. $\forall k \quad p(k) \rightarrow P(k + 1)$

EXAMPLE:

Use Mathematical Induction to prove that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} \quad \text{for all integers } n \geq 1$$

SOLUTION:

Let $P(n) : 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$

1. Basis Step:

$P(1)$ is true.

For $n = 1$, left hand side of $P(1)$ is the sum of all the successive integers starting at 1 and ending at 1, so LHS = 1 and RHS is

$$R.H.S = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

so the proposition is true for $n = 1$.

2. Inductive Step: Suppose $P(k)$ is true for, some integers $k \geq 1$.

$$(1) \quad 1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$$

To prove $P(k + 1)$ is true. That is,

$$(2) \quad 1 + 2 + 3 + \cdots + (k+1) = \frac{(k+1)(k+2)}{2}$$

Consider L.H.S. of (2)

$$\begin{aligned} 1 + 2 + 3 + \cdots + (k+1) &= 1 + 2 + 3 + \cdots + k + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \quad \text{using (1)} \\ &= (k+1) \left[\frac{k}{2} + 1 \right] \\ &= (k+1) \left[\frac{k+2}{2} \right] \end{aligned}$$

$$\begin{aligned} &= (k+1) \frac{k+2}{2} \\ &= \frac{(k+1)(k+2)}{2} = \text{RHS of (2)} \end{aligned}$$

Hence by principle of Mathematical Induction the given result true for all integers greater or equal to 1.

EXERCISE:

Use mathematical induction to prove that
 $1+3+5+\dots+(2n-1) = n^2$ for all integers $n \geq 1$.

SOLUTION:

Let $P(n)$ be the equation $1+3+5+\dots+(2n-1) = n^2$

1. Basis Step:

$P(1)$ is true

For $n = 1$, L.H.S of $P(1) = 1$ and

R.H.S = $1^2 = 1$

Hence the equation is true for $n = 1$

2. Inductive Step:

Suppose $P(k)$ is true for some integer $k \geq 1$. That is,

$$1 + 3 + 5 + \dots + (2k - 1) = k^2 \dots\dots\dots(1)$$

To prove $P(k+1)$ is true; i.e.,

$$1 + 3 + 5 + \dots + [2(k+1)-1] = (k+1)^2 \dots\dots\dots(2)$$

Consider L.H.S. of (2)

$$\begin{aligned} 1 + 3 + 5 + \dots + [2(k+1)-1] &= 1 + 3 + 5 + \dots + (2k+1) \\ &= 1 + 3 + 5 + \dots + (2k-1) + (2k+1) \\ &= k^2 + (2k+1) \quad \text{using (1)} \\ &= (k+1)^2 \\ &= \text{R.H.S. of (2)} \end{aligned}$$

Thus $P(k+1)$ is also true. Hence by mathematical induction, the given equation is true for all integers $n \geq 1$.

EXERCISE:

Use mathematical induction to prove that
 $1+2+2^2+\dots+2^n = 2^{n+1}-1$ for all integers $n \geq 0$

SOLUTION:

Let $P(n)$: $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$

1. Basis Step:

$P(0)$ is true.

For $n = 0$

L.H.S of $P(0) = 1$

R.H.S of $P(0) = 2^{0+1} - 1 = 2 - 1 = 1$

Hence $P(0)$ is true.

2. Inductive Step:

Suppose $P(k)$ is true for some integer $k \geq 0$; i.e.,

$$1+2+2^2+\dots+2^k = 2^{k+1} - 1 \dots \dots \dots (1)$$

To prove $P(k+1)$ is true, i.e.,

$$1+2+2^2+\dots+2^{k+1} = 2^{k+1+1} - 1 \dots \dots \dots (2)$$

Consider LHS of equation (2)

$$\begin{aligned} 1+2+2^2+\dots+2^{k+1} &= (1+2+2^2+\dots+2^k) + 2^{k+1} \\ &= (2^{k+1} - 1) + 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+1+1} - 1 = \text{R.H.S of (2)} \end{aligned}$$

Hence $P(k+1)$ is true and consequently by mathematical induction the given propositional function is true for all integers $n \geq 0$.

EXERCISE:

Prove by mathematical induction

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for all integers $n \geq 1$.

SOLUTION:

Let $P(n)$ denotes the given equation

1. Basis step:

$P(1)$ is true

For $n = 1$

L.H.S of $P(1) = 1^2 = 1$

$$\begin{aligned} \text{R.H.S of } P(1) &= \frac{1(1+1)(2(1)+1)}{6} \\ &= \frac{(1)(2)(3)}{6} = \frac{6}{6} = 1 \end{aligned}$$

So L.H.S = R.H.S of $P(1)$. Hence $P(1)$ is true

2. Inductive Step:

Suppose $P(k)$ is true for some integer $k \geq 1$;

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6} \dots \dots \dots (1)$$

To prove $P(k+1)$ is true; i.e.;

$$1^2 + 2^2 + 3^2 + \dots + (k+1)^2 = \frac{(k+1)(k+1+1)(2(k+1)+1)}{6} \dots (2)$$

$$\begin{aligned}
 1^2 + 2^2 + 3^2 + \dots + (k+1)^2 &= 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 \\
 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\
 &= (k+1) \left[\frac{k(2k+1)}{6} + (k+1) \right] \\
 &= (k+1) \left[\frac{k(2k+1) + 6(k+1)}{6} \right] \\
 &= (k+1) \left[\frac{2k^2 + k + 6k + 6}{6} \right] \\
 &= (k+1) \left[\frac{2k^2 + 7k + 6}{6} \right] \\
 &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\
 &= \frac{(k+1)(k+2)(2k+3)}{6} \\
 &= \frac{(k+1)(k+1+1)(2(k+1)+1)}{6}
 \end{aligned}$$

EXERCISE:

Prove by mathematical induction

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1} \quad \text{for all integers } n \geq 1$$

SOLUTION:

Let P(n) be the given equation.

1. Basis Step:

P(1) is true

For n = 1

$$\text{L.H.S of } P(1) = \frac{1}{1 \cdot 2} = \frac{1}{1 \times 2} = \frac{1}{2}$$

$$\text{R.H.S of } P(1) = \frac{1}{1+1} = \frac{1}{2}$$

Hence P(1) is true

2. Inductive Step:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

To prove P(k+1) is true. That is

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(k+1)(k+1+1)} = \frac{k+1}{(k+1)+1} \dots \dots \dots (2)$$

Now we will consider the L.H.S of the equation (2) and will try to get the R.H.S by using equation (1) and some simple computation.

Consider LHS of (2)

$$\begin{aligned} & \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(k+1)(k+2)} \\ &= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2)+1}{(k+1)(k+2)} \\ &= \frac{k^2+2k+1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} \\ &= \frac{k+1}{k+2} \\ &= \text{RHS of (2)} \end{aligned}$$

Hence $P(k+1)$ is also true and so by Mathematical induction the given equation is true for all integers $n \geq 1$.

EXERCISE:

Use mathematical induction to prove that

$$\sum_{i=1}^{n+1} i2^i = n \cdot 2^{n+2} + 2, \quad \text{for all integers } n \geq 0$$

SOLUTION:

1. Basis Step:

To prove the formula for $n = 0$, we need to show that

$$\sum_{i=1}^{0+1} i \cdot 2^i = 0 \cdot 2^{0+2} + 2$$

$$\text{Now, L.H.S} = \sum_{i=1}^1 i \cdot 2^i = (1)2^1 = 2$$

$$\text{R.H.S} = 0 \cdot 2^2 + 2 = 0 + 2 = 2$$

Hence the formula is true for $n = 0$

2. Inductive Step:

Suppose for some integer $n = k \geq 0$

$$\sum_{i=1}^{k+1} i \cdot 2^i = k \cdot 2^{k+2} + 2 \quad \cdots \cdots \cdots (1)$$

We must show that $\sum_{i=1}^{k+2} i \cdot 2^i = (k+1) \cdot 2^{k+2} + 2$

$$\cdots \cdots \cdots (2)$$

Consider LHS of (2)

$$\begin{aligned}
 \sum_{i=1}^{k+2} i \cdot 2^i &= \sum_{i=1}^{k+1} i \cdot 2^i + (k+2) \cdot 2^{k+2} \\
 &= (k \cdot 2^{k+2} + 2) + (k+2) \cdot 2^{k+2} \\
 &= (k+k+2)2^{k+2} + 2 \\
 &= (2k+2) \cdot 2^{k+2} + 2 \\
 &= (k+1)2 \cdot 2^{k+2} + 2 \\
 &= (k+1) \cdot 2^{k+1+2} + 2 \\
 &= \text{RHS of equation (2)}
 \end{aligned}$$

Hence the inductive step is proved as well. Accordingly by mathematical induction the given formula is true for all integers $n \geq 0$.

EXERCISE:

Use mathematical induction to prove that

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n} \quad \text{for all integers } n \geq 2$$

SOLUTION:

1. Basis Step:

For $n = 2$

$$\text{L.H.S} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{R.H.S} = \frac{2+1}{2(2)} = \frac{3}{4}$$

Hence the given formula is true for $n = 2$

2. Inductive Step:

Suppose for some integer $k \geq 2$

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k} \quad \dots \dots \dots (1)$$

We must show that

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{(k+1)^2}\right) = \frac{(k+1)+1}{2(k+1)} \dots \dots \dots (2)$$

Consider L.H.S of (2)

$$\begin{aligned}
 & \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{(k+1)^2}\right) \\
 &= \left(\frac{1}{1} \cdot \frac{1}{1} \cdot \frac{1}{1} \cdots \frac{1}{1}\right) \left(\frac{1}{2^2} \cdot \frac{1}{3^2} \cdots \frac{1}{(k+1)^2}\right) \\
 &= \left(\frac{k+1}{2k}\right) \left(\frac{1}{1 - \frac{1}{(k+1)^2}}\right) \\
 &= \left(\frac{k+1}{2k}\right) \left(\frac{(k+1)^2}{(k+1)^2 - 1}\right) \\
 &= \left(\frac{1}{2k}\right) \left(\frac{k^2 + 2k + 1}{k^2 + 2k + 1 - 1}\right) \\
 &= \frac{k^2 + 2k}{2k(k+1)} = \frac{k(k+2)}{2k(k+1)} \\
 &= \frac{k+1+1}{2(k+1)} = \text{RHS of (2)}
 \end{aligned}$$

Hence by mathematical induction the given equation is true

EXERCISE:

Prove by mathematical induction

$$\sum_{i=1}^n i(i!) = (n+1)! - 1 \quad \text{for all integers } n \geq 1$$

SOLUTION:

1. Basis step:

For $n = 1$

$$\text{L.H.S} = \sum_{i=1}^1 i(i!) = (1)(1!) = 1$$

$$\begin{aligned}
 \text{R.H.S} &= (1+1)! - 1 = 2! - 1 \\
 &= 2 - 1 = 1
 \end{aligned}$$

Hence

$$\sum_{i=1}^1 i(i!) = (1+1)! - 1$$

which proves the basis step.

2. Inductive Step:

Suppose for any integer $k \geq 1$

$$\sum_{i=1}^k i(i!) = (k+1)! - 1 \quad \dots\dots\dots(1)$$

We need to prove: $\sum_{i=1}^{k+1} i(i!) = (k+1+1)! - 1$

Consider LHS of (2)

$$\begin{aligned}
 \sum_{i=1}^{k+1} i(i!) &= \sum_{i=1}^k i(i!) + (k+1)(k+1)! && \text{Using (1)} \\
 &= (k+1)! - 1 + (k+1)(k+1)! \\
 &= (k+1)! + (k+1)(k+1)! - 1 \\
 &= [1 + (k+1)](k+1)! - 1 \\
 &= (k+2)(k+1)! - 1 \\
 &= (k+2)! - 1 \\
 &= \text{RHS of (2)}
 \end{aligned}$$

Hence the inductive step is also true.

Accordingly, by mathematical induction, the given formula is true for all integers $n \geq 1$.

